

CHEM 121

Experiment 1: Measurements and Density Determination

Introduction:

Being capable of properly recording observations and data and performing calculations with collected data are very important skills that a chemist must possess.

All measurements contain systematic errors and random errors. **Systematic errors** from flaws in the equipment or design of the experiment and result in inaccuracy. **Accuracy** can be described as the agreement of an experimental value with the true or accepted value. If systematic errors are present, the recorded value will always be higher than the true value or always be lower than the true value. Poor laboratory technique is not considered to be a source of systematic error as this is something that is easily controlled. **Random errors** affect precision (reproducibility) and are associated with multiple, independent measurements that are expected to be the same. Random errors have an equal chance of being either positive or negative.

The **precision** associated with a measurement or calculated result can be indicated by the number of significant figures. For a measurement, the number should include **all digits known with certainty and the first uncertain (estimated) digit**. For single measurements, the first digit for which you are uncertain is generally the one resulting from estimating to 1/10th of the smallest scale division. Using different measuring devices may allow measurements to be determined with a greater degree of precision. As the number of significant figures increase, the more certain (precise) is the measurement.

Whenever possible, scientists will choose to make at least three measurements (expecting the same answer each time). However, due to random errors, the three numbers may be slightly different. In order to report the most realistic value based on three pieces of data, the average or mean (experimental value) may be reported. If we want to know the accuracy of this result, we need to know the true or accepted value. If the experimental value appears to be different from the accepted value, one or more errors are present in the measurement. To quantify the size of these errors (even though we do not know the source(s) of these errors), we can do so in terms of **absolute and relative (percent) error**:

$$\text{Absolute error} = \text{Experimental value} - \text{Accepted value}$$

$$\text{Relative error} = \frac{\text{Experimental value} - \text{Accepted value}}{\text{Accepted value}} \times 100\%$$

Procedure:

This experiment consists of four parts (A, B, C, and D) set up at stations around the lab, which can be completed in any order. Your lab instructor will assign each student to a station to begin and will then cycle each student around the lab at timed intervals.

Part A:

At Station A you will find a setup of different graduated measuring instruments for measuring volumes. Read the volume and record the reading to the correct number of significant figures for each instrument.

Part B:

At Station B you are provided with unknown samples labelled A, B and C. Choose only one sample and record your sample letter in the space provided.

1. Complete the following steps to weigh your chosen sample by taring on the top-loading balance:
 - a. Place a weighing boat on the top-loading balance after cleaning the pan.
 - b. Press briefly on the tare (zero) button and make sure the display changes to zero. The weight of the boat is now tared.
 - c. With a spatula, carefully add the sample to the weighing boat on the balance to weigh between 1.50 g to 2.00 g of the sample.
 - d. Record the mass of the sample.
 - e. Keep the sample in the weighing boat for part 2.
2. Complete the following steps to weigh your sample from part 1 by difference on the analytical balance:
 - a. With all the doors of the balance closed (and nothing on the balance), press briefly on the zero button and make sure the display changes to zero.
 - b. Now place a clean and dry 50 mL beaker on the balance and close all the doors. Record the mass of the empty beaker.
 - c. Remove the beaker from the balance and put it on the benchtop. Carefully transfer the sample from the weighing boat (from part 1) into the beaker.
 - d. Place the beaker with the sample back into the balance and close all the doors. Record the mass of the beaker with the sample.

Part C:

At Station C you will find the equipment required to measure the density of an unknown sample. Density is a physical property and can be used to identify substances. Density is defined as the mass per unit of volume. You are provided with unknown samples labelled 1, 2 and 3. Choose only one sample and record your sample number in the space provided. Complete the following steps to gather the necessary data:

- a. On an analytical balance, weigh a clean and dry 50 mL Erlenmeyer flask + stopper.
Record the mass of the flask + stopper.
- b. Remove the flask from the balance, remove the stopper, and set them on a clean benchtop.
- c. Pipette 25.00 mL of sample and carefully transfer the sample into the flask and stopper it.
- d. Reweigh the flask + stopper with the sample. Record the mass of the flask + stopper with the sample.
- e. Measure the temperature of the sample.

Part D:

At Station D you will find a setup of different substances. Record observations for each of the provided substances.